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BRIEF NOTE

Pre-exercise hyperventilation can significantly increase performance in the 50-meter front crawl



Une hyperventilation pré-exercice peut significativement améliorer les performances lors du 50-mètre crawl

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Summary

Objectives. – The aim of this study was to measure the effect of a 30-second pre-exercise maximal voluntary hyperventilation on the performance during a 50 m front crawl.

Material and methods. – Nine well-trained swimmers (5 males [21.0 ± 8.5 years, 181.4 ± 3.5 cm, 71.2 ± 3.9 kg] and 4 females [21.0 ± 8.7 years, 166.2 ± 5.0 cm, 55.2 ± 2.2 kg]) performed a 50 m front crawl sprint either in normal conditions (NO) or after hyperventilation (HV) (6 maximal breathing cycles realized in 30 seconds followed by 30 seconds of rest).

Results. – Average velocity for the 50 m front crawl was significantly higher after HV ($1.81 \pm 0.13 \text{ m} \cdot \text{s}^{-1}$ vs. $1.79 \pm 0.14 \text{ m} \cdot \text{s}^{-1}$, $P < 0.01$). As a result, performance improves ($27.79 \pm 2.01 \text{ s}$ vs. $28.08 \pm 2.17 \text{ s}$, $P < 0.01$). The number of breathing cycles recorded during each race was significantly lower under HV compared to NO (1.88 ± 0.92 and 2.66 ± 1.41 , $P < 0.01$). Moreover, the first air intake was significantly delayed under HV conditions (29.55 ± 8.67 vs. $23.55 \pm 6.10 \text{ m}$, $P < 0.01$). The stroke rate was slightly increased under HV conditions while stroke length was kept nearly constant.

Conclusion. – A pre-exercise maximal voluntary hyperventilation can significantly increase performance on the 50 m front crawl in well-trained swimmers.

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MOTS CLÉS

Sprint ;
Fatigue ;
Nage ;
Alcalose respiratoire

Résumé

Objectifs. – Dans cette étude, nous cherchons à vérifier si une hyperventilation pré-exercice volontaire maximale de 30 secondes peut induire une augmentation de la performance sur 50 m crawl.

Matériel et méthodes. – Neuf nageurs entraînés (5 garçons [$21,0 \pm 8,5$ ans, $181,4 \pm 3,5$ cm, $71,2 \pm 3,9$ kg] et 4 filles [$21,0 \pm 8,7$ ans, $166,2 \pm 5,0$ cm, $55,2 \pm 2,2$ kg]) ont effectué un sprint de 50 m crawl, soit dans des conditions normales (NO), soit après une hyperventilation (HV) (6 cycles respiratoires d'amplitude maximale de 30 secondes suivis par 30 secondes de récupération passive).

Résultats. – La vitesse moyenne lors du 50 m crawl est significativement supérieure dans les conditions HV par rapport aux conditions NO ($1,81 \pm 0,13$ m.s⁻¹ vs $1,79 \pm 0,14$, $p < 0,01$). Par conséquent, la performance est supérieure dans les conditions HV ($27,79 \pm 2,01$ s vs $28,08 \pm 2,17$ s, $p < 0,01$). Le nombre de cycles de respiration enregistrés lors de chaque course est significativement inférieur dans les conditions HV par rapport aux conditions NO ($1,88 \pm 0,92$ et $2,66 \pm 1,41$, $p < 0,01$). De plus, la première prise d'air est apparue significativement plus loin dans les conditions HV par rapport aux conditions NO ($29,55 \pm 8,67$ vs $23,55 \pm 6,10$ m, $p < 0,01$). La fréquence des cycles de bras a été légèrement augmentée dans les conditions HV alors que la longueur des cycles est restée constante.

Conclusion. – Une hyperventilation maximale volontaire augmente significativement la vitesse de nage lors d'un 50 m crawl.

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1. Introduction

In swimming, many coaches advice their athletes to voluntarily hyperventilate before sprint events. This pre-exercise hyperventilation is supposed to saturate the blood with oxygen and remove carbon dioxide, leading to what is called respiratory alkalosis, which induce an increase of blood pH. This pre-exercise increase in blood pH can delay the rapid decrease in blood pH induced by sprint exercise. Many studies using pre-exercise supplementation with bicarbonate ions to increase pre-exercise blood pH and buffer capacity have reported significantly higher performance compared to control [1]. More recently, some authors also reported that sodium bicarbonate ingestion positively affected repeated swim sprint performance [1]. However, some others do not report any significant improvements in performance after NaHCO₃ ingestion [1]. The dosage size, the timing of ingestion and the exercise type seems to influence the results. Moreover, bicarbonate ions may cause gastrointestinal upset in ~50% of individuals. Voluntary hyperventilation can also lead to an increase of blood pH (respiratory alkalosis). Indeed, a brief hyperventilation (30 seconds of voluntary maximal hyperventilation [6 maximal breathing cycles as deep as possible in 30 seconds]) reduced significantly the partial pressure of end-tidal carbon dioxide (PETCO₂) values below 30 mmHg, which suggest a significant increase of blood pH [2]. In fact, such a decrease (10 mmHg) in PETCO₂ values corresponds to an increase of 0.08 pH units. This increase is comparable to that reported by some studies using bicarbonate supplementation [1]. Thus, the aim of this study was to evaluate the effect of a 30-second maximal voluntary hyperventilation followed by 30 seconds of passive rest on 50 m crawl performance. It was hypothesized that under this condition the increase in blood pH due to HV could have a positive effect on performance in two ways:

- reduction of fatigue by inducing a pre-exercise respiratory alkalosis;
- decreasing the number of breathing cycles during the race, as air uptake is known to significantly modify front crawl technique and decrease velocity [3].

2. Material and methods

In this study, all subjects performed two 50 m front crawl sprints, one in normal conditions (NO) and one after a brief voluntary hyperventilation (HV) followed by 30 seconds of passive rest. The two 50 m front crawl sprints were performed on the same day, separated by 30 minutes of rest. This recovery time was considered sufficient since the swimmers were well trained and used to participating in competitions where the delay between races is about 15 minutes. The two sprints were completed in a randomized order after a 15-minute competition-like warm-up. A few microliters of arterialized capillary blood were drawn to determine blood lactate immediately after and 3 minutes after each sprint. Blood lactate concentration was determined enzymatically using a lactate analyzer (Microzym, Cetrix, France). Finally, to avoid any mistakes concerning the timekeeping of the race, some biomechanical aspects were evaluated to accurately determine the number of breathing cycles, the distance covered before the first air intake, and the number of arms cycles. Four HD waterproof camcorders (VPC-WH1, Sanyo) were employed as depicted in Fig. 1: one camcorder was placed under water, and three were placed outside of the water.

2.1. Subjects

Five male (21.0 ± 8.5 years, 181.4 ± 3.6 cm, 71.2 ± 4.0 kg) and 4 female (21.0 ± 8.7 years, 166.2 ± 5.1 cm, 55.2 ± 2.2 kg) members of a first-division swimming club

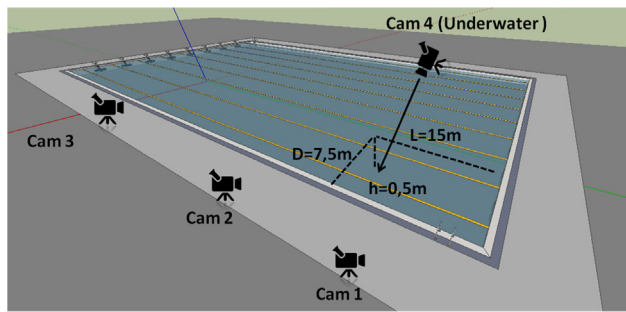


Figure 1 Placement of the camcorders.

volunteered to participate to this study. All subjects had trained 6 times per week (10 hours per week) for 5 years minimum. They were all informed of the protocol and possible risks associated with the study and all gave written informed consent. The protocol was approved by the ethics committee of the university of Balamand (Lebanon).

2.2. Hyperventilation protocol

In this study the hyperventilation protocol (HV) consisted of 6 maximal respiratory cycles in 30 seconds (2 seconds for a maximal inspiration followed by 3 seconds of maximal expiration) [2], but introduced a 30-second passive rest between the end of the hyperventilation and the test.

2.3. Biomechanical analysis

All the tests were filmed using 4 HD waterproof camcorders (VPC-WH1, Sanyo). A side-view of the swimmer's motion was video recorded using a 50 Hz HD waterproof camcorder (VPC-WH1, Sanyo). This camcorder was located as shown in Fig. 1. The specific positioning of this camcorder aimed at evaluating the effect of hyperventilation on spatiotemporal parameters for the last meters of the race. These camcorders enabled us to calculate the average stroke rate (from hand entry at the first stroke to hand entry at the second stroke) and the index of coordination, in accordance with the protocol outlined by Chollet et al. [4].

Three external side-view camcorders (VPC-WH1, Sanyo), synchronized with the underwater side-view camera, filmed all trials laterally. These camcorders measured the following parts of the 50 m race: camcorder 1, from 0 to 15 m; camcorder 2, from 15 m to 35 m; and camcorder 3, from 35 m to 50 m. The stroke rate was obtained by counting the requisite number of video frames for three strokes of the distance covered. The stroke length (SL) was calculated from the average velocity (V) and the stroke rate (SR): $SL = V/SR \times 60$.

The index of coordination is defined as the time gap between the beginning of propulsion in the first right arm stroke and the end of propulsion in the first left arm stroke, and between the beginning of propulsion in the second left arm stroke and the end of propulsion in the first right arm stroke [1,2]. For each trial, the average index of coordination was calculated for three complete strokes and expressed as a percentage of the mean duration of the stroke. When a lag time occurred between the propulsive phases of the two arms, the stroke coordination was called

Table 1 Performance, number of breathing and arms cycles, and biomechanical data determined during the 50 m front crawl sprint, as measured in normal conditions (NO) or after hyperventilation (HV).

	NO	HV
25 m time (s)	12.89	12.67**
	0.95	0.85
50 m time (s)	28.08	27.79**
	2.17	2.01
Average velocity	1.83	1.79
	0.13	0.14
Stroke rate (stroke.min ⁻¹)	49.10	50.20*
	8.5	8.8
Stroke length (m)	2.18	2.15
	0.35	0.42
Index of coordination (%)	0.9	1.1*
	0.3	0.3
Breathing cycles	2.66	1.88**
	1.41	0.92
Distance covered before the first breathing cycle (m)	23.55	29.77**
	6.10	8.67
Lactate concentration (mmol.L ⁻¹) at the end	4.15	3.99
	1.69	1.44
Lactate concentration (mmol.L ⁻¹) after 3'	8.02	8.42
	1.60	1.33

Values are means $\pm$ standard deviation.

* Significant difference: $P < 0.05$.

** Significant difference: $P < 0.01$.

'catch-up' (IdC < 0%). When the propulsive phase of one arm started when the other arm ended its propulsive phase, the coordination was called 'opposition' (IdC = 0%). When the propulsive phases of the two arms overlapped, the coordination was called 'superposition' (IdC > 0%).

2.4. Statistical analysis

Data are expressed as means $\pm$ standard deviation (SD). Comparisons between the two conditions (HV and NO) were made using a paired *t*-test. A value of $P < 0.05$ was accepted as the minimal level of statistical significance. Correlation analysis was performed by a Spearman rank test for non-normally distributed variables and by a Pearson test for normally distributed variables. All statistical analyses were performed using statistical software (SIGMA STAT for windows, version 3.11, 2004).

3. Results

Performances and data determined during the two 50 m front crawl sprints are presented in Table 1. Average velocity was significantly higher under HV conditions than under NO ($1.81 \pm 0.13 \text{ m.s}^{-1}$ vs. 1.79 ± 0.14 , $P < 0.01$, for HV and NO respectively). This corresponded to an increase of $1.03\% \pm 0.82$ (range -0.21% to $+2.25\%$). As a result, performance improved ($27.79 \pm 2.01 \text{ s}$ vs. $28.08 \pm 2.17 \text{ s}$,

$P < 0.01$, for HV and NO respectively). The number of breathing cycles recorded during each race was significantly lower during the HV conditions compared to normal conditions (1.88 ± 0.92 and 2.66 ± 1.41 , $P < 0.01$, for HV and NO respectively). This corresponded to a decrease $24.80 \pm 20.67\%$ (range 0 to 50%). The first air intake was significantly delayed during the HV conditions (29.55 ± 8.67 vs. 23.55 ± 6.10 m, $P < 0.01$, for HV and NO respectively). This corresponded to an increase of 26% (range 0 to 53%) representing 6.22 ± 4.41 m (range 0 to 12 m).

The stroke rate was slightly increased under HV conditions, while stroke length was kept nearly constant. Finally, the index of coordination was slightly increased and positive for both conditions, which corresponds to superposition coordination ($\text{IdC} > 0\%$).

4. Discussion

This study shows that a short pre-exercise hyperventilation followed by a 30-second passive recovery increased significantly the velocity of a 50 m front crawl. In this study, the average velocity during the first and the second 25 m were significantly higher under HV conditions, and event performance was significantly higher. Indeed, only one subject did not perform better under HV conditions (-0.21%). This result cannot be mainly explained by an increase in the anaerobic pathway output, since lactate concentrations determined immediately after or 3 minutes after the sprint were comparable under both conditions. In contrast, the time of first air intake and the distance covered before the first air intake are significantly different. In this study, the number of breathing cycles was significantly lower in HV conditions compared to normal conditions. This helps to explain the increased performance observed in HV conditions, since several studies reported that velocity decreased significantly during air uptake [3]. In the same way, the distance covered before the first air uptake is significantly higher under HV conditions (29.77 ± 8.67 vs. 23.55 ± 6.10 , $P < 0.01$). These results suggest that the improved performance reported under HV conditions is mainly due to HV's important effect on the urge to breathe. This can be explained as HV has been shown to induce a significant 10 mmHg decrease in PETCO_2 that can reflect a significantly decrease in arterial CO_2 partial pressure (respiratory alkalosis) [2].

Finally, after HV, the first air intake appeared 6.22 ± 4.41 m after under HV conditions. This is in accordance with many previous results reporting that a pre-apnea hyperventilation can significantly increase time of apnea both at rest and during heavy exercise.

From a biomechanical point of view, the results show that stroke rate was slightly increased under HV conditions. This increase in stroke rate may be mainly explained since arm swimming is easier without breathing than while breathing [3]. At the same time, stroke length was kept nearly constant, which enables mean velocity to increase. Index of coordination was slightly increased and positive for both conditions, which correspond to superposition coordination ($\text{IdC} > 0\%$). As previously depicted in Chollet et al. [4], an increase in IdC is related to high velocity. Our results therefore suggest an increase in velocity due to higher propulsive forces generated during superposition action under HV conditions.

5. Conclusion

In conclusion, the present study showed that a brief pre-exercise voluntary maximal hyperventilation could significantly increase performance in the 50 m front crawl in well-trained swimmers. This study points the advantages of brief pre-exercise HV. More studies are now required to explain our finding more clearly. Further studies should focus on measuring more physiological parameters such as blood pH (after the HV, after the test) and the swimmers' technique throughout the race to better understand the effect of HV on fatigue. All this data should allow to improve the use of HV (duration, number of cycles, recovery time), which may contribute to even more improvements in sprint performance.

Disclosure of interest

The authors declare that they have no conflicts of interest concerning this article.

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